CYBERSECURITY INTERNSHIP

WEEK 10 – INTERNSHIP REPORT

SOC – WAZUH SETUP & ALERT MONITORING

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Week	Week 10
SIN	DG/SEPTEMBER/CYBER/254
Submitted to	DG INTERNS HUB

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1. INTRODUCTION

A Security Operations Center (SOC) plays an important role in an organization's cybersecurity infrastructure. It provides centralized security monitoring, threat detection, incident response, and investigation.

Security Information and Event Management (SIEM) solutions collect, correlate, and analyze security logs from multiple sources. This helps identify suspicious activities and possible cyber threats.

Wazuh is an open-source security monitoring platform. It supports endpoint security monitoring, log analysis, file integrity monitoring, vulnerability detection, and security event analysis. Integrating Wazuh with an intrusion detection system such as Snort can improve threat visibility and security event correlation.

2. OBJECTIVES

- Understand the fundamental concepts of Security Operations Center (SOC) operations and SIEM technologies.
- Study the architecture, components, and deployment requirements of the Wazuh security monitoring platform.
- Explore endpoint monitoring, agent installation, configuration, and connectivity with the Wazuh server.
- Understand security event collection, log analysis, alert generation, and threat investigation.
- Examine File Integrity Monitoring (FIM) and vulnerability detection features provided by Wazuh.
- Explore the integration of Snort IDS alerts with Wazuh for centralized monitoring and improved threat detection.

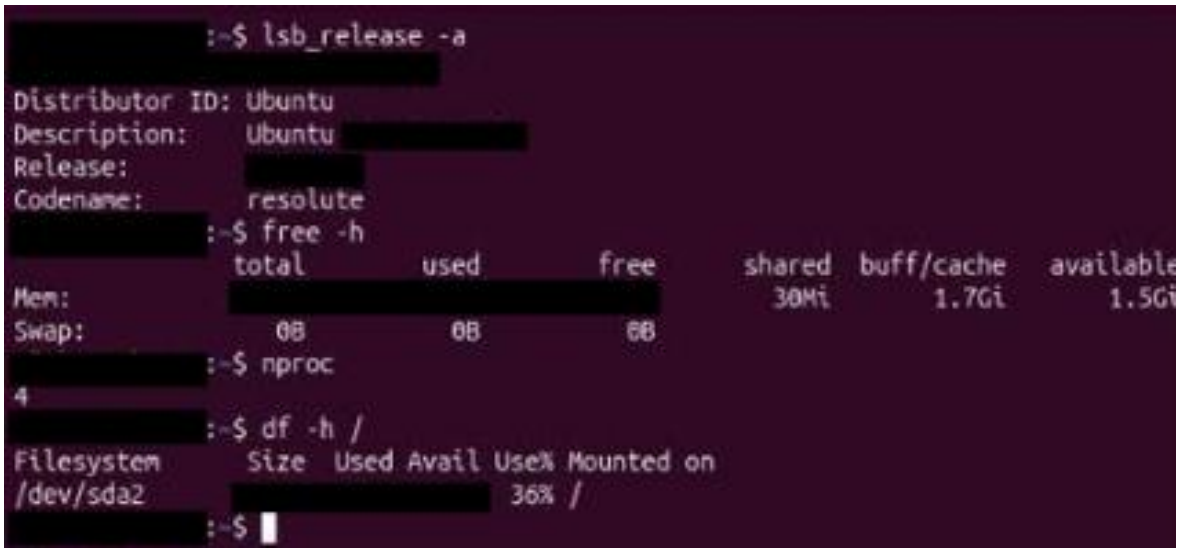


Figure 1. Visual from the internship presentation – objectives and SOC monitoring.

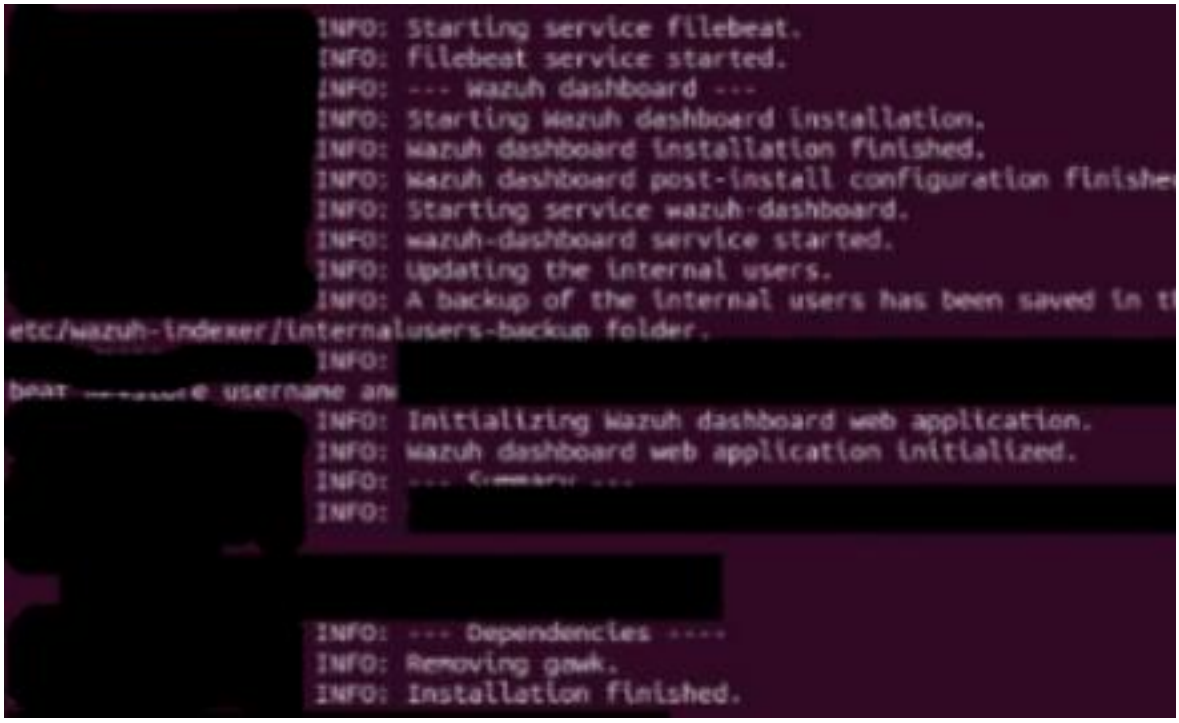


Figure 2. Visual from the internship presentation – objectives and SOC monitoring.

3. TOOLS AND TECHNOLOGIES USED

Wazuh SIEM: Used for centralized security monitoring, log analysis, and threat detection.

Linux Operating System: Provides the platform for installing, configuring, and testing security tools.

Wazuh Manager and Agent: Enable endpoint monitoring, event collection, and security analysis.

Wazuh Indexer and Dashboard: Provide security data storage, visualization, and alert investigation.

Snort IDS: Monitors network traffic and detects suspicious activities using predefined rules.

Virtual Machine Environment: Provides an isolated setup for deploying and testing security tools.

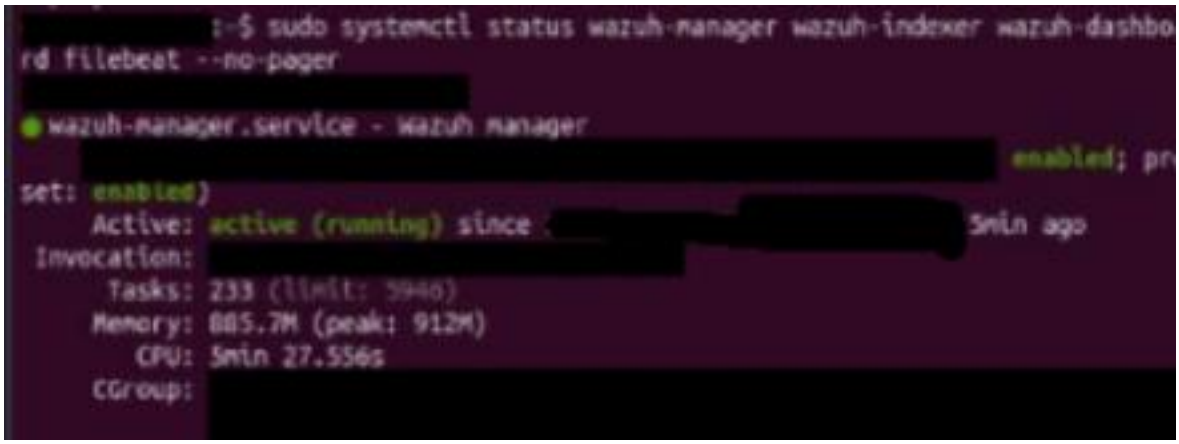


Figure. Tool or technology reference included in the original presentation.



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4. WAZUH SIEM ARCHITECTURE

The Wazuh SIEM architecture consists of four main components:

Wazuh Agent: Collects security events, system logs, and monitoring data from endpoints.

Wazuh Manager: Processes incoming events and identifies threats using decoders and detection rules.

Wazuh Indexer: Stores and organizes security data for efficient searching and analysis.

Wazuh Dashboard: Provides an interface for security monitoring, alert visualization, and investigation.

Monitoring Workflow

Endpoint System → Wazuh Agent → Wazuh Manager → Wazuh Indexer → Wazuh Dashboard

This architecture provides centralized security monitoring, improves threat visibility, and helps security analysts detect and investigate suspicious activities across monitored endpoints.

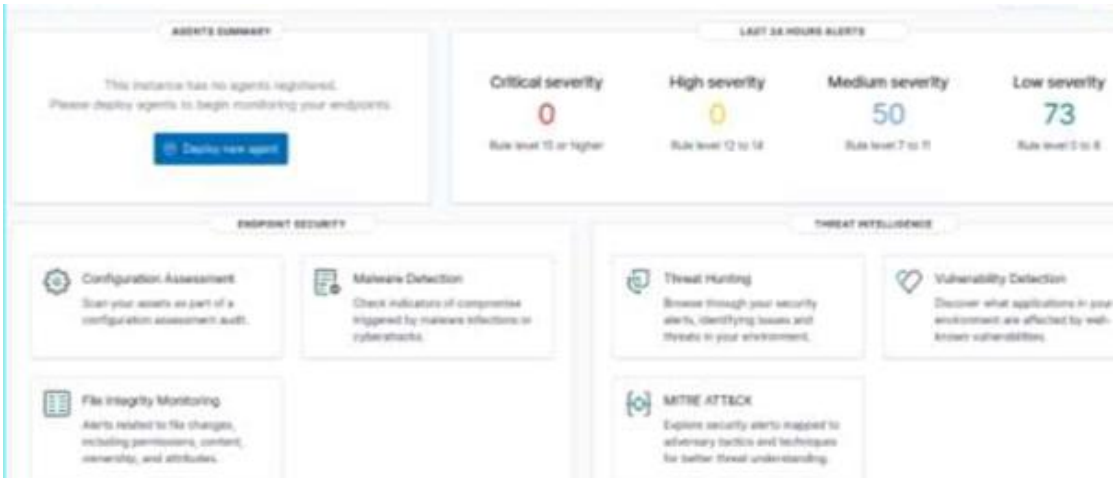


Figure. Wazuh SIEM architecture and monitoring workflow visual.

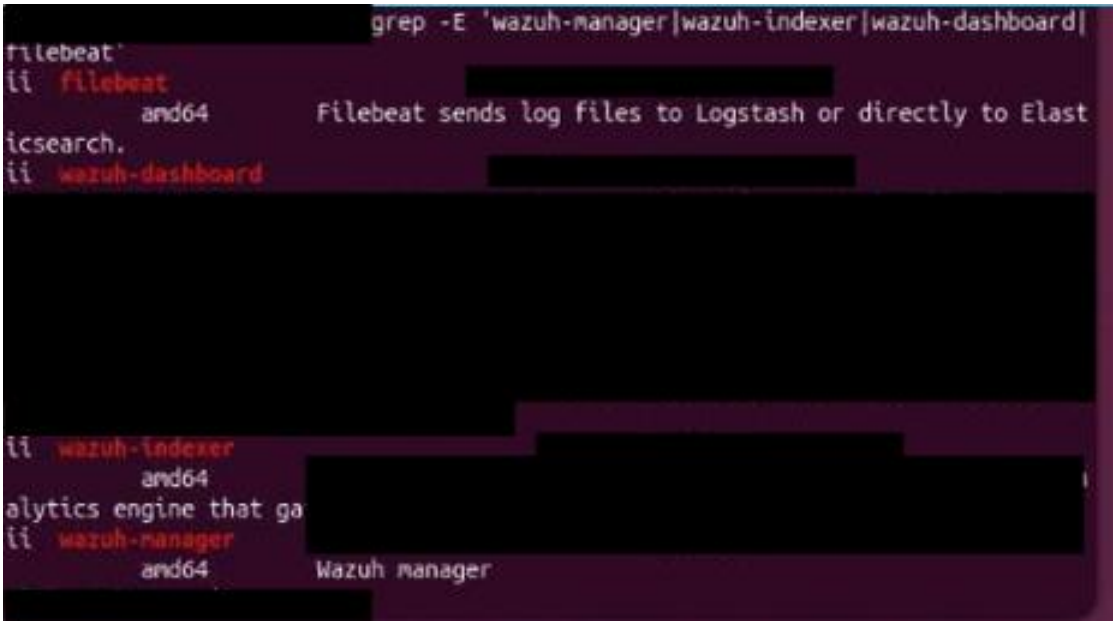


Figure. Wazuh SIEM architecture and monitoring workflow visual.

5. ENDPOINT MONITORING AND AGENT CONFIGURATION

The endpoint monitoring and agent configuration process described in the presentation is:

1. Prepare and configure the endpoint system.
2. Download and install the Wazuh agent.
3. Set the Wazuh manager IP address and enrollment details.
4. Start and enable the agent service.
5. Check connectivity between the agent and manager.
6. Verify agent registration and successful event collection.

Monitoring Benefits

- Provides centralized monitoring of endpoint systems.
- Collects system logs and authentication events.
- Detects suspicious activities and unauthorized changes.
- Helps security analysts investigate potential threats.
- Improves overall endpoint security and visibility.



Figure. Endpoint monitoring and Wazuh agent configuration visual.

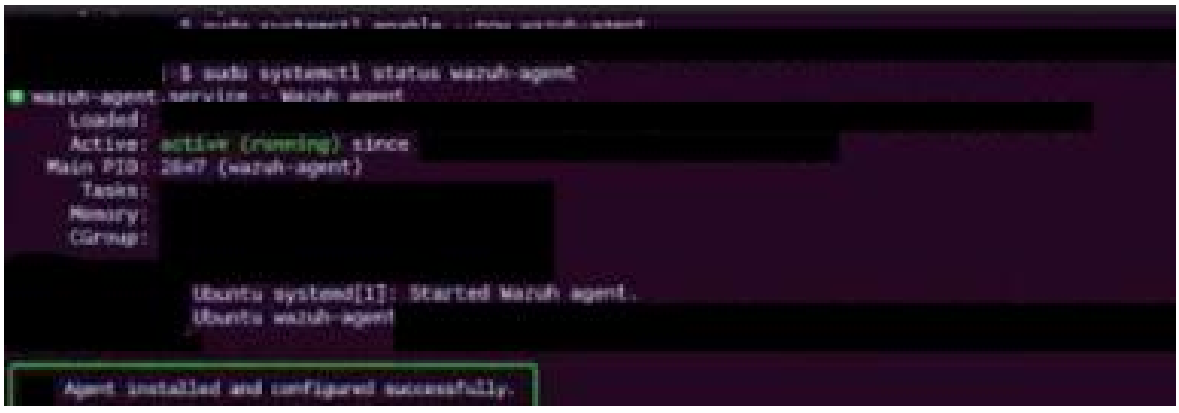


Figure. Endpoint monitoring and Wazuh agent configuration visual.

6. SECURITY EVENT AND ALERT MONITORING

The presentation covers the following key event categories and alert investigation activities:

Authentication Monitoring

- Monitors successful and failed login attempts.
- Identifies suspicious login patterns and unauthorized access.

File Integrity Monitoring (FIM)

- Detects file creation, modification, and deletion.
- Helps identify unauthorized changes.

Alert Investigation

- Reviews security alerts and affected endpoints.
- Analyzes events to identify potential threats.

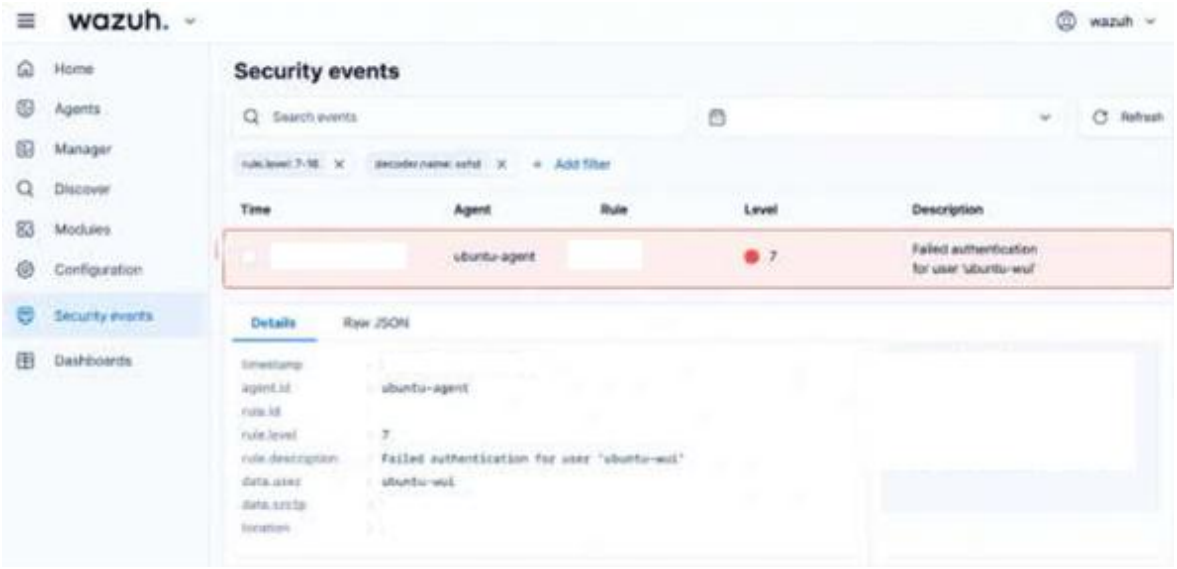


Figure. Security event monitoring and alert-analysis visual.



Figure. Security event monitoring and alert-analysis visual.

7. SNORT IDS AND WAZUH INTEGRATION

Snort IDS monitors network traffic and detects suspicious activities using predefined rules. Snort alerts can be forwarded to Wazuh for centralized monitoring and analysis.

Integration Workflow

7. Snort monitors incoming and outgoing network traffic.
8. Suspicious activity generates alerts.
9. Alerts are stored in log files.
10. Wazuh collects and analyzes the Snort logs.
11. SOC analysts review the generated alerts.

Security Benefits

- Centralized monitoring of network and endpoint events.
- Improved threat detection and visibility.
- Faster investigation of suspicious activities.
- Better incident analysis and reporting.



Figure. Snort IDS and Wazuh integration visual from the presentation.

8. CHALLENGES ENCOUNTERED

- Faced difficulties during Wazuh installation and dashboard configuration.
- Experienced issues with agent connectivity and service verification.
- Could not fully confirm test alert generation and collection.

9. LEARNING OUTCOMES

- Understood SOC operations and SIEM architecture.
- Learned about Wazuh components and endpoint monitoring.
- Gained knowledge of authentication monitoring and file integrity monitoring.
- Studied alert analysis and Snort integration with Wazuh.

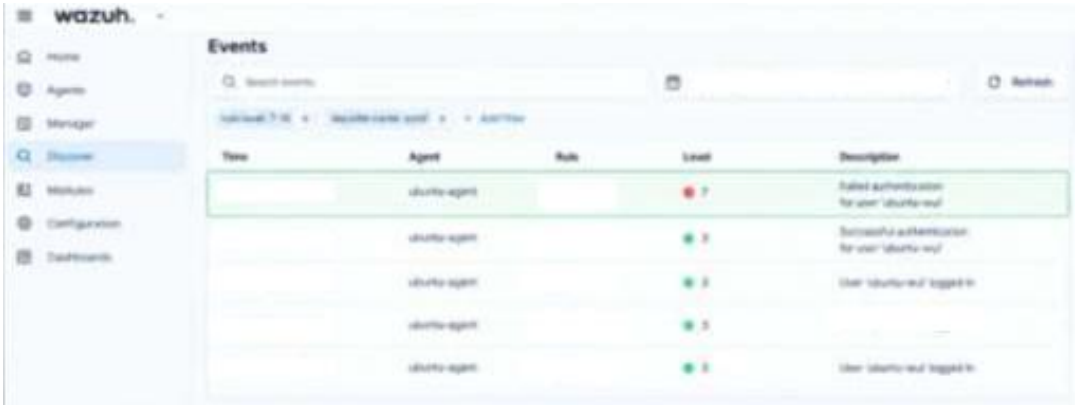


Figure. Supporting visual included in the Week 10 presentation.

10. CONCLUSION

The Week 10 activities provided a basic understanding of SOC operations, SIEM concepts, and Wazuh security monitoring. Despite some configuration challenges, the study helped build knowledge of endpoint monitoring, event collection, threat detection, and alert investigation.